

Structural Sufficiency, Sacrificial Pruning, and Minimal Throughput

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Abstract

Recent results across neuroscience, machine learning, and systems theory converge on a counterintuitive principle: functional coherence does not scale with material mass or parameter count, but with the preservation of a minimal structural core. This paper formalizes that principle by unifying three empirical domains—extreme hydrocephalus, lottery-ticket phenomena in neural networks, and biological pruning mechanisms—under the framework of the Theory of Axiomatic Necessity (TNA). We argue that sustained coherence requires active sacrifice of non-contributing substructures to remain above a minimal functional threshold F_{min} . This process is operational, energy-constrained, and measurable via negentropic throughput Ψ . The result is a non-metaphysical theory of structural sufficiency with implications for neuroscience, artificial intelligence, and minimal-resource learning.

1. Introduction

Classical intuitions across disciplines equate functionality with quantity: more neurons, more parameters, more components imply greater capability. Yet multiple independent lines of evidence now contradict this assumption. A human brain reduced to a thin cortical mantle can sustain ordinary life; neural networks pruned to 10% of their parameters can match baseline performance; biological systems routinely destroy vast internal structure to survive.

This paper proposes that these are not anomalies but manifestations of a single structural law: *coherence persists only if a system actively sheds non-essential structure while preserving its axiomatic core*. We formalize this claim using the TNA framework and introduce sacrificial pruning as a necessary dynamic of survival.

The Theory of Axiomatic Necessity establishes that no time-evolving system can maintain coherence using only internally derivable rules. At least one non-derivable constraint F_{min} is required to prevent cumulative incoherence.

Crucially, this constraint cannot be static. In real systems, coherence must be *maintained*, not merely postulated. This leads to an operational formulation: the enforcement of F_{min} requires continuous entropy export. The measurable footprint of this enforcement is the system's negentropic throughput Ψ , expressed in W/K.

A detailed operational and thermodynamic grounding of Ψ is provided in *From Axiomatic Necessity (F_{min}) to Measurable Throughput (Ψ)* (Bresciano, 2026), which we adopt here as the formal definition of system-level sustainability.

3. Sacrificial Dynamics in Biological Systems

Biology offers multiple canonical examples of sacrificial pruning:

- **Apoptosis** eliminates cells that threaten organism-level coherence.
- **Synaptic pruning** removes redundant or noisy connections during development.
- **Immune clonal deletion** destroys self-reactive lymphocytes to preserve systemic identity.

In all cases, survival depends not on maximal retention but on selective destruction. Systems that fail to prune collapse under their own internal load. This establishes sacrifice not as a defect, but as an active stability mechanism.

4. Extreme Hydrocephalus: Structural Sufficiency Without Mass

A striking illustration is the well-documented case of a French man with massive congenital hydrocephalus, retaining approximately 10–20% of normal brain volume. Despite this, he lived a socially and professionally normal life, with only mildly reduced IQ scores.

This case falsifies any theory of consciousness or cognition that depends on:

- a critical mass of neurons,
- fixed neuroanatomical localization,
- or uniform distribution of function.

Instead, it supports a model in which the remaining tissue reorganizes around a minimal sufficient structure. The observable brain ($\mathcal{N}_0$) is drastically reduced, yet coherence persists due to latent structural support ($\mathcal{N}_1$) realized through lifelong plasticity.

The brain did not compensate by *working harder*—it survived by never attempting to sustain more structure than strictly necessary.

5. Lottery Tickets and Structural Minima in Machine Learning

The Lottery Ticket Hypothesis demonstrates that dense neural networks contain sparse subnetworks capable of matching full-model performance when trained in isolation. This is often framed as a sparsity result, but structurally it is a discovery of F_{min} in artificial systems.

Training identifies a minimal parameter configuration that preserves function; pruning removes the dissipative periphery. However, unlike biological systems, most artificial neural networks possess static connectivity and limited capacity for continuous structural adaptation. Pruning typically occurs *after* training, not during it.

This highlights a critical asymmetry:

- Biological systems prune continuously and plastically.
- Machine learning systems prune discretely and retrospectively.

The difference is not conceptual, but operational.

6. Implications for Minimal-Resource Machine Learning

The convergence between extreme biological reduction and winning-ticket phenomena suggests a general strategy for minimal-resource ML: prioritize preservation of dominant functional modes rather than exhaustive parameter coverage.

From the TNA perspective, efficient models do not approximate full representations with fewer parameters; they converge toward structural sufficiency. The preserved modes correspond to dominant eigenvectors of the system's effective connectivity operator, shaped by boundary conditions and task-driven constraints.

This reframes efficiency as a stability problem rather than a compression problem. Systems fail not when they lack capacity, but when incoherence exceeds their throughput Ψ . Models designed to operate near F_{min} minimize energy expenditure while retaining function.

Importantly, this argument does not depend on claims about consciousness or agency. It applies to any system required to maintain coherent input–output behavior under resource constraints.

7. Discussion: Fractality, Energy Minimization, and the Trunk Principle

Across domains, pruning follows a consistent geometry: peripheral, high-variance structures are removed first; invariant cores persist. This is the logic of fractal systems under energy minimization.

One prunes the leaves, not the trunk.

The trunk encodes the generative structure; leaves dissipate energy. Systems that attempt to preserve everything exhaust their throughput and collapse. Systems that sacrifice aggressively converge toward F_{min} .

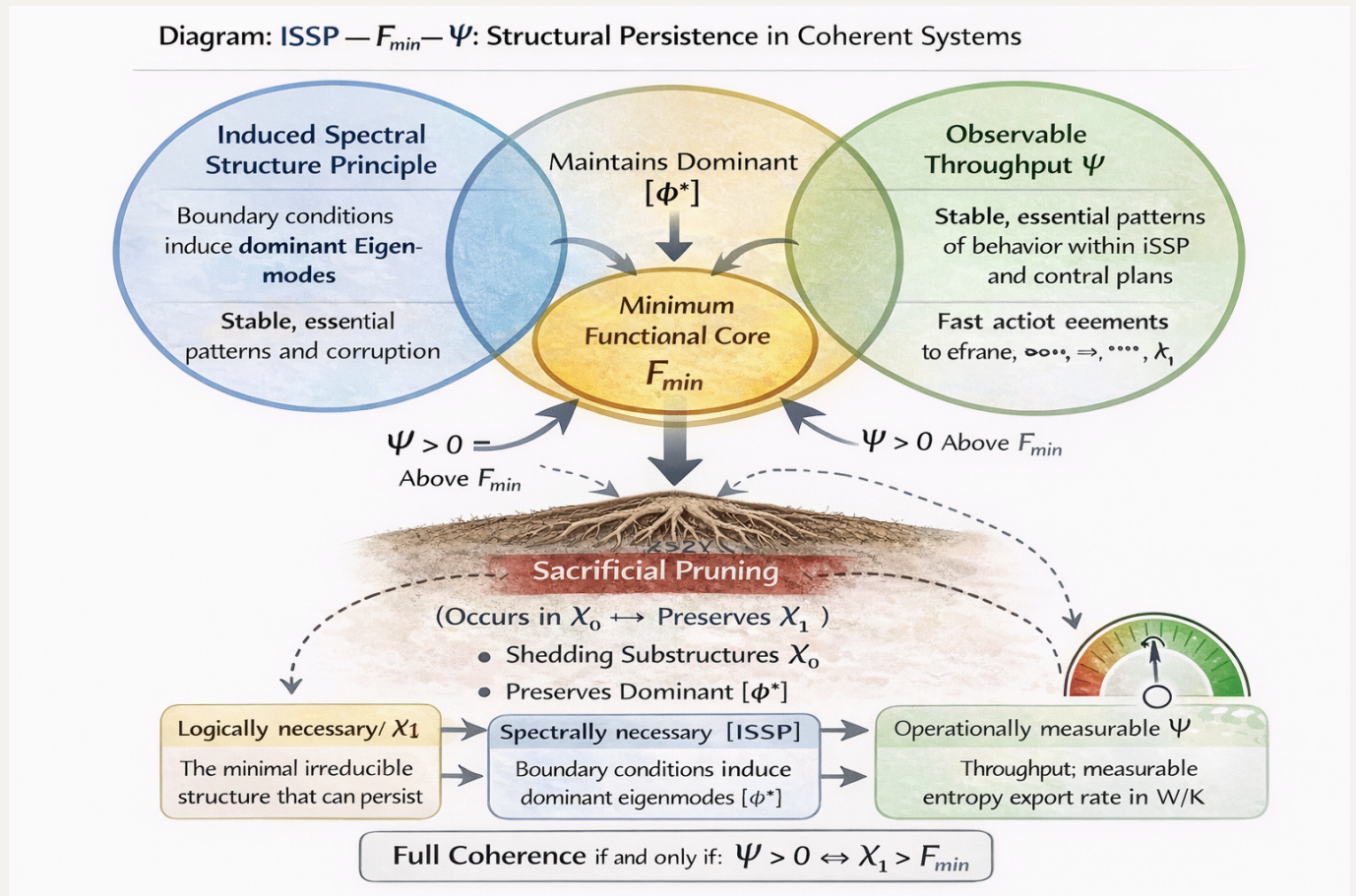
8. Conclusion

The systems examined here did not survive despite massive reduction. They survived *because of it*.

The brain with 10% of its mass, the neural network with 10% of its parameters, and the organism that deletes millions of its own cells all exemplify the same law: coherence persists only when non-contributing structure is actively shed.

The brain did not fail with 90% less material. It triumphed because it never attempted to be more than strictly necessary.

ISSP– $F_{\min}$ – Ψ Diagram



Explanation of the ISSP– $F_{\min}$ – Ψ Diagram

The diagram summarizes how **structural persistence** is maintained in coherent systems across biological, computational, and physical domains by linking three concepts: **ISSP**, $F_{\min}$, and Ψ .

1. ISSP: Boundary-Induced Structure (Left)

The **Induced Spectral Structure Principle (ISSP)** states that **boundary conditions select a small set of dominant eigenmodes** that govern system behavior.

These modes are not arbitrary: they arise because the system is constrained by physical, biological, or informational boundaries.

- In brains: cranial volume, sensory input, metabolic limits.
- In neural networks: architecture, loss function, data interface.
- In physical systems: geometry, dissipation channels, conservation laws.

ISSP explains why **most degrees of freedom are spectrally irrelevant**: they correspond to subdominant modes that can be removed without destroying functionality.

2. $F_{\min}$: The Minimal Functional Core (Center)

At the intersection lies $F_{\min}$ — the **minimum functional core** required for the system to remain coherent.

$F_{\min}$ is **not a quantity of material**, neurons, or parameters.

It is a **structural invariant**: the smallest set of modes, connections, or operations that can still implement the system's defining function.

- In the hydrocephalic brain, this core survives as a thin cortical mantle.
- In machine learning, it appears as a “winning ticket”.
- In living systems, it corresponds to the trunk rather than the leaves.

Crucially, **which components constitute $F_{\min}$ is not fixed** — only the *existence* of such a core is necessary.

3. Ψ : Observable Throughput (Right)

Ψ represents **measurable operational throughput**: entropy export, task completion, behavioral output, or information processing capacity.

- $\Psi > 0$ indicates that the system is actively sustaining coherence.
- When Ψ drops below a critical threshold, the system collapses regardless of remaining mass or complexity.

Importantly, Ψ is **a consequence**, not a cause:
it depends on whether the system's structure remains above $F_{\min}$.

4. Sacrificial Pruning: Why Reduction Preserves Function

The lower part of the diagram introduces **sacrificial pruning**.

This pruning:

- Occurs in the **observable domain** (X_0): excess neurons, parameters, cells, protocols.
- Preserves the **necessary domain** (X_1): dominant modes selected by ISSP.

Examples include:

- Apoptosis in biology.
- Synaptic pruning in development.
- Clonal deletion in immune systems.
- Hardware-aware sparsity in ML.
- Massive cortical loss with preserved cognition in extreme hydrocephalus.

The key insight is that **pruning is not loss** but **energy minimization under structural constraints**.

By removing non-contributing substructures, the system prevents Ψ from collapsing below viability.

5. The Core Equivalence Condition

At the bottom, the diagram states the formal condition:

$$\text{Full coherence if and only if } \Psi > 0 \iff X_1 \geq F_{\min}$$

This means:

- A system survives **only if** its necessary structure remains intact.
 - Excess structure is optional; insufficient structure is fatal.
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6. Conceptual Unification

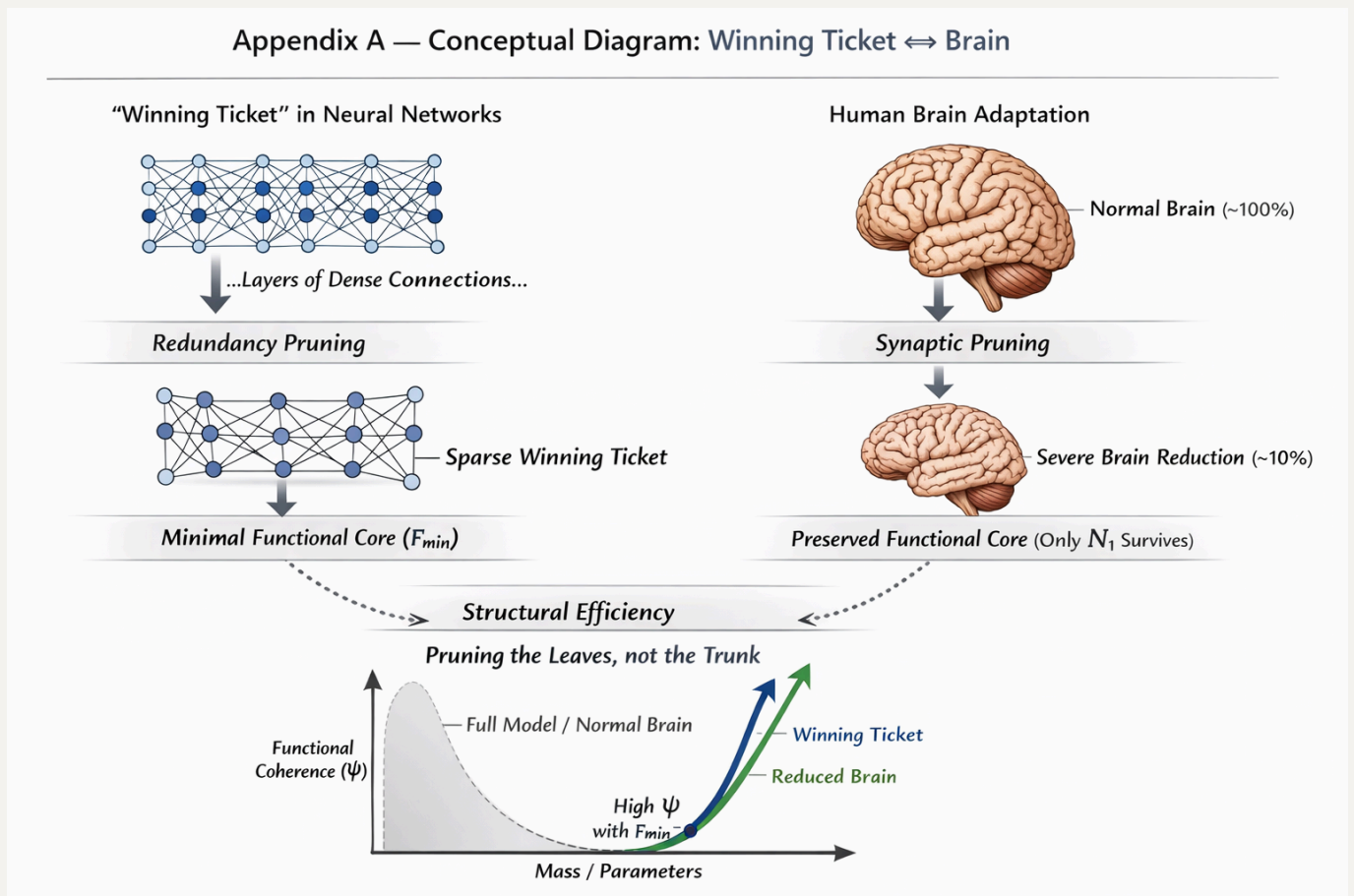
The diagram closes the technical circle:

- **ISSP** explains *why* a small core exists.
- $F_{\min}$ defines *what* must be preserved.
- Ψ shows *whether* the system is operational.
- **Sacrificial pruning** explains *how* systems remain viable under constraint.

In this view, the French hydrocephalus case and lottery ticket networks are not anomalies — they are **canonical examples of structural sufficiency**

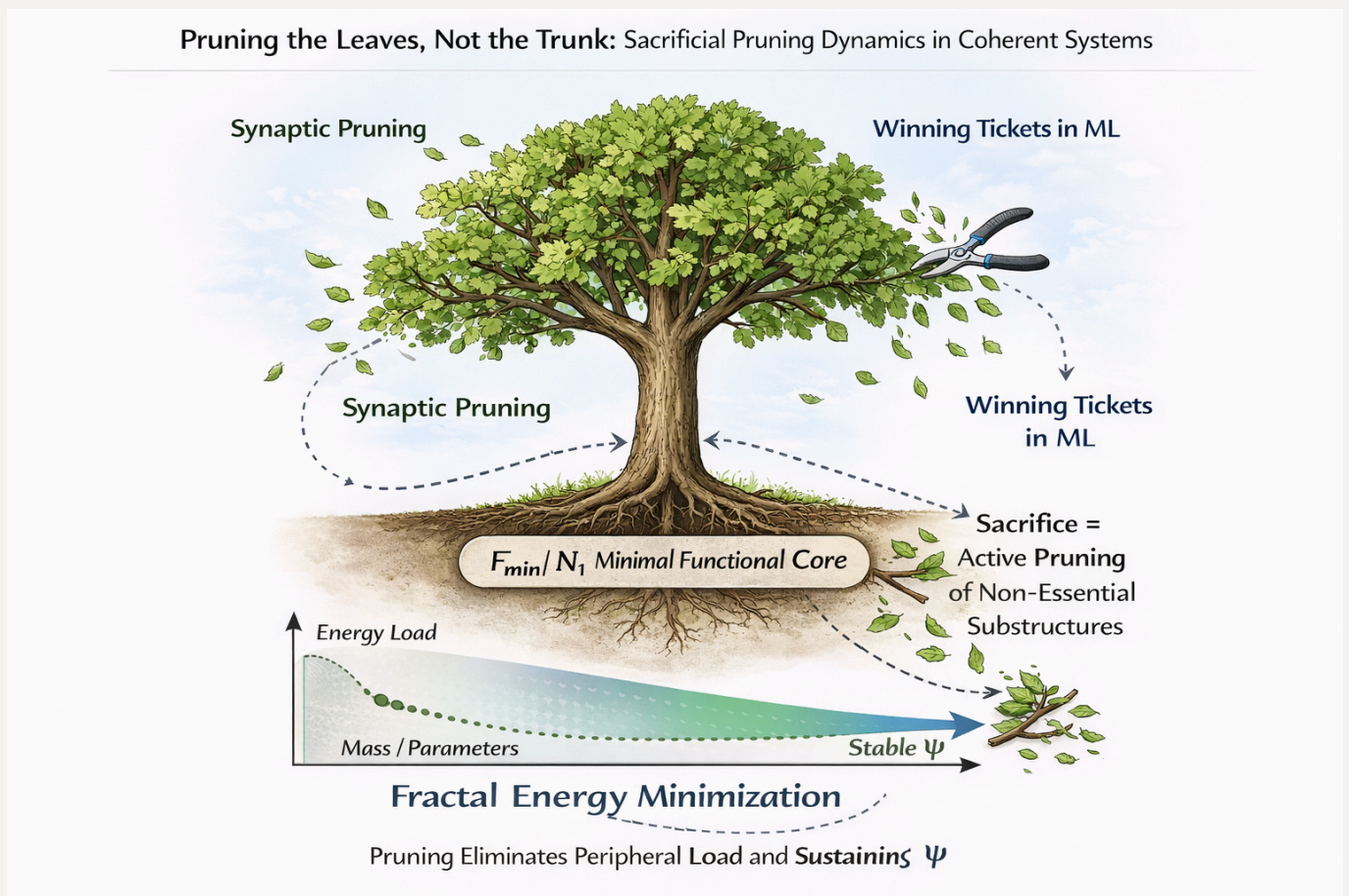
A system does not survive by preserving everything it has, but by preserving only what it cannot lose.

Appendix A — Conceptual Diagram: Winning Ticket $\leftrightarrow$ Brain



The diagram illustrates parallel convergence toward F_{min} in biological and artificial systems, emphasizing continuous plastic pruning versus discrete post-training pruning, and the role of throughput Ψ in sustaining coherence over time.

“Sacrificial Pruning” Diagram



Explanation of the “Sacrificial Pruning” Diagram (Trunk / Leaves $\leftrightarrow F_{\min} / \mathcal{N}_1$)

The diagram represents a general principle of **structural sufficiency under resource constraints**, using a tree as a unifying metaphor across biology, neuroscience, and machine learning.

1. Trunk vs. Leaves: Structural Core and Peripheral Load

- **The trunk** represents the **minimal functional core** of a system, labeled explicitly as $F_{\min} / \mathcal{N}_1$.
 - This is the smallest irreducible structure that preserves coherence, identity, and functionality.
 - Damage to the trunk implies systemic collapse.
- **The leaves and peripheral branches** represent **redundant, non-essential, or**

energy-dissipative substructures.

- They enhance performance under abundance but are not required for survival or coherence.

The key claim visualized is simple but strong:

Systems do not fail when mass is reduced; they fail only when the minimal structural core is violated.

2. Sacrificial Pruning as an Active Process

The pruning shears symbolize **active, not passive, reduction**.

This pruning corresponds to:

- **Apoptosis** in biology
- **Synaptic pruning** in neurodevelopment
- **Clonal deletion** in the immune system
- **Parameter pruning / winning tickets** in machine learning

In all cases, **non-contributing substructures are deliberately removed** to reduce energy load and preserve global coherence (Ψ).

This is why the diagram labels pruning explicitly as **sacrifice**:

- Something is lost,
 - But the loss is *structurally necessary* to remain above $F_{\min}$.
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3. Energy Minimization and Fractal Sufficiency

The lower part of the diagram shows:

- A decreasing curve of **energy load vs. mass/parameters**
- Stability (Ψ) improving as peripheral load is removed

This captures a **principle of fractal energy minimization**:

- The trunk contains, in compressed form, the full functional logic of the system.

- Removing leaves does not destroy function because they are *fractal repetitions*, not axiomatic supports.

Thus, pruning does not simplify *randomly* — it reveals the **true invariant**.

4. Winning Tickets and the Brain Case

The diagram explicitly links:

- **Winning tickets in ML** (minimal subnetworks that retain full performance)
- **Extreme hydrocephalus cases** (functional cognition with drastically reduced brain volume)

In both:

- The surviving structure corresponds to $F_{\min}$
- Function persists because **organization, not mass**, is decisive

This challenges models that assume:

- A critical amount of hardware,
 - Or fixed anatomical localization,
as necessary conditions for cognition or intelligence.
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5. Conceptual Takeaway

The diagram encodes the central thesis visually:

“Pruning the leaves, not the trunk”
is the universal strategy by which coherent systems survive under constraint.

Or, more formally:

Coherent systems remain viable by actively shedding peripheral structures while preserving their axiomatic core ($F_{\min} / \mathcal{N}_1$). Energy minimization reveals structure, not absence.

Objections and Clarifications: Why This Is Not Merely Sparsity, Plasticity, or Occam's Razor

A natural reaction to the framework presented here is to interpret it as a rebranding of familiar ideas: sparsity in machine learning, plasticity in neuroscience, or Occam's razor in philosophy of science. While superficially related, these interpretations miss the central structural claim. This section clarifies the distinctions.

1. Not Just Sparsity

Sparsity describes a *statistical property*: many parameters are unused or weakly contributing, and can therefore be removed without significant loss of performance.

The present framework makes a stronger, **structural** claim:

- It is not that *many parameters are unnecessary*,
- but that *a minimal irreducible structure must exist* for coherence to persist.

Winning tickets are not valuable because they are sparse, but because they instantiate $F_{\min}$ — the smallest substructure that preserves functional closure. Sparsity can be a consequence of approaching $F_{\min}$, but it is not the explanation.

A sparse system without the correct structural core collapses; a dense system that preserves $F_{\min}$ remains coherent. Thus, sparsity is neither sufficient nor necessary — structural sufficiency is.

2. Not Just Plasticity

Neural plasticity explains *how* functions can relocate or reorganize after damage or during development. It does not explain *why* cognition persists under extreme structural reduction.

In the extreme hydrocephalus case, plasticity alone cannot account for the outcome:

- Plasticity redistributes function,
- but redistribution requires a **stable invariant** to redistribute *around*.

The framework introduced here identifies that invariant as $F_{\min}$. Plasticity operates *above* this threshold; it cannot replace it. Below $F_{\min}$, no amount of plasticity restores coherence.

Thus, plasticity is an adaptive mechanism, not a foundational principle. It presupposes the existence of a minimal viable structure.

3. Not Occam’s Razor

Occam’s razor is a **methodological heuristic**: prefer simpler explanations when competing theories explain the same observations.

The present claim is not about explanatory preference, but about **ontological and operational necessity**.

$F_{\min}$ is not “simpler because we like simplicity”; it is enforced by:

- energy minimization,
- functional closure,
- and the avoidance of infinite regress or incoherence.

A system that fails to preserve its minimal core does not merely become less elegant — it ceases to function. This is a constraint, not a choice.

Occam’s razor selects among theories.

$F_{\min}$ constrains which systems can exist at all.

4. Summary Distinction

CONCEPT	WHAT IT EXPLAINS	WHAT IT MISSES
Sparsity	Parameter efficiency	Structural necessity
Plasticity	Functional redistribution	Minimal coherence threshold
Occam’s razor	Theory selection	System viability constraints
This work	Structural sufficiency ($F_{\min}$)	—

5. Core Claim Restated

The framework does not argue that systems *prefer* to be minimal.

It argues that systems **must** preserve a minimal irreducible structure to remain coherent.

Reduction succeeds only when it reveals the trunk.

Reduction fails when it cuts into it.

That distinction — between optional complexity and axiomatic necessity — is the central contribution of this work.

References

- Feuillet, L. et al. (2007). *Brain of a white-collar worker*. The Lancet.
- Frankle, J., & Carbin, M. (2019). *The Lottery Ticket Hypothesis*. ICLR.
- Bresciano, C. (2026). *From Axiomatic Necessity (F_{min}) to Measurable Throughput (Ψ)*. Zenodo.
- Prigogine, I. (1977). *Self-Organization in Nonequilibrium Systems*.
- Landauer, R. (1961). *Irreversibility and Heat Generation in the Computing Process*.

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